

09/15/2025: Quantifiers

CSCI 246: Discrete Structures

Textbook reference: Sec. 11, Scheinerman

Announcements

- Reminder: If you finish the quiz early, please be silent until everyone else is finished (so as to not distract students who are still thinking).
- Reminder: Mike's office hours are canceled until he returns from out of town Friday 9/25. Madhav's office hours and SmartyCats tutoring are running as normal. Feel free to contact Mike by email if you'd like.

Today's Agenda

- Weekly Quiz (≈ 15 mins)
- Review group exercises (≈ 10 mins)
- Mini-lecture (≈ 15 mins)
- Group exercises (≈ 10 mins)

Weekly Quiz #3

1. (Sec. 7 – *Boolean Algebra*.) DeMorgan's laws are:

$$\neg(x \wedge y) = (\neg x) \vee (\neg y) \quad \text{and} \quad \neg(x \vee y) = (\neg x) \wedge (\neg y).$$

Prove the first of these (using truth tables). For reference: $\neg$ means “not”, $\wedge$ means “and”, and $\vee$ means “or”.

2. (Sec. 8 – *Lists*.) We have a club with 20 members. We want to elect an executive board consisting of a president, a vice president, a secretary, and a treasurer. How many ways can we do this (assuming no member of the club can fill two offices)?
3. (Sec. 9 – *Factorial*.) Your friend Sandy has 6 different books of classical literature, 7 different *romantasy* (i.e. romance fantasy) books, and 8 different self-help books. (a) In how many different ways can Sandy's books be arranged on a bookshelf? (b) In how many different ways can Sandy's books be arranged on the bookshelf if all books of the same category are grouped together?

Subtopics from today's reading

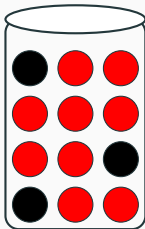
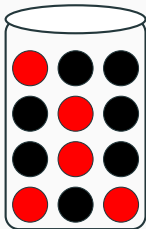
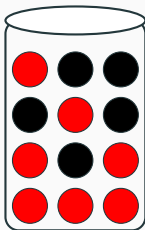
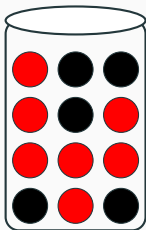
Scheinerman Sec. 11 (Quantifiers) covers the following subtopics

1. Using quantified statements
 - a. How to interpret quantified statements
 - b. How to negate quantified statements
2. Proving quantified statements
 - a. Proving existential statements
 - b. Proving universal statements

Outline for Mini-Lecture:

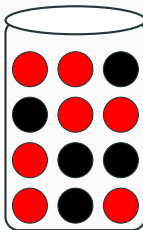
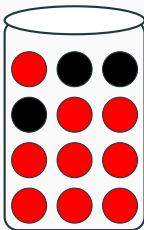
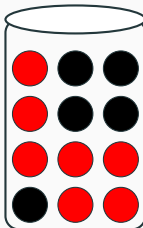
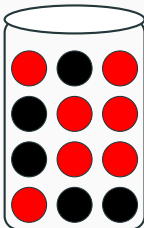
- Using quantified statements
- Proving quantified statements
- Applications

Proposition. $\forall \text{ jar} \in \text{jars}, \exists \text{ ball} \in \text{jar} : \text{ball is red}$



Poll. What is the proposition saying in English?

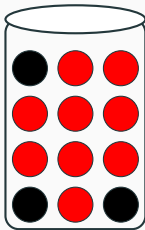
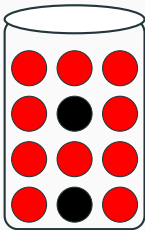
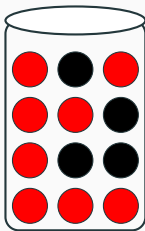
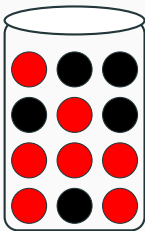
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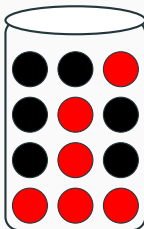
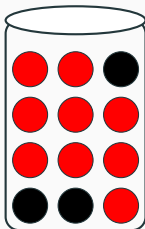
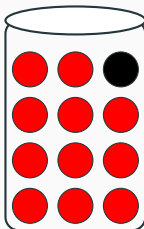
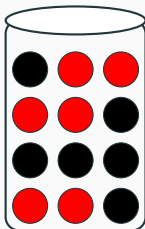
Solution. For each jar in the set of jars, there is a ball in the jar such that the ball is red.

Proposition. $\forall \text{ jar} \in \text{jars}, \exists \text{ ball} \in \text{jar} : \text{ball is red}$



Poll. Is this proposition true or false?

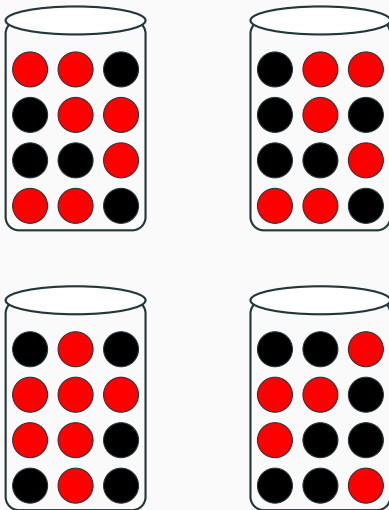
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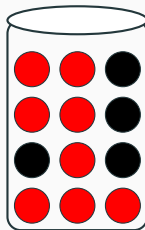
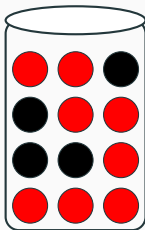
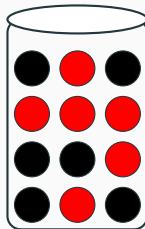
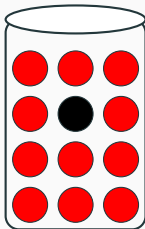
Solution. True

Proposition. $\forall \text{ jar} \in \text{jars}, \exists \text{ ball} \in \text{jar} : \text{ball is red}$



Poll. How can we change the picture to make the proposition false?

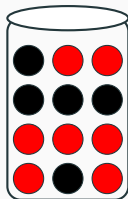
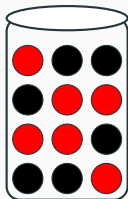
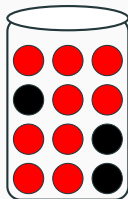
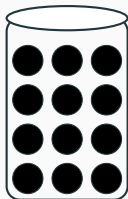
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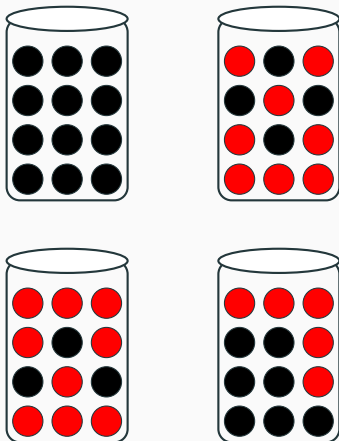
Solution. We need to change one of the jars so that it has no red balls.

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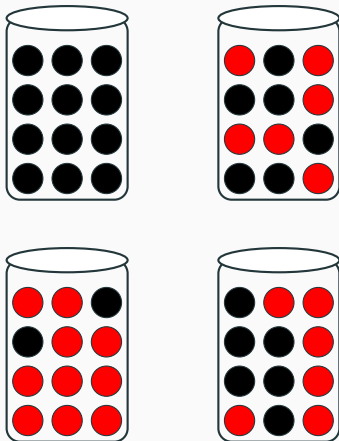
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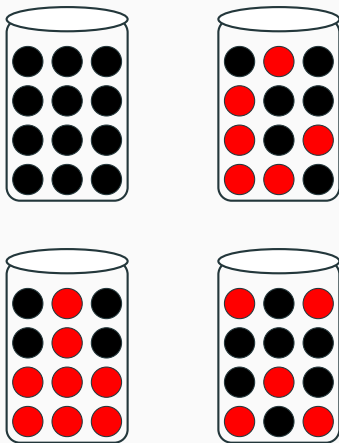
Solution. For each jar in the set of jars, there is a ball in the jar such that the ball is red.

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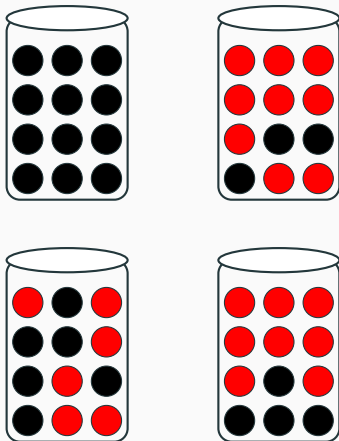
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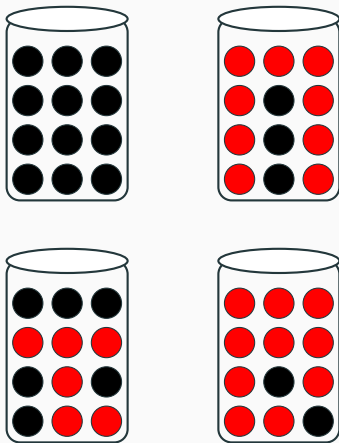
Solution. False.

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Poll. How can we change the proposition to make it true?

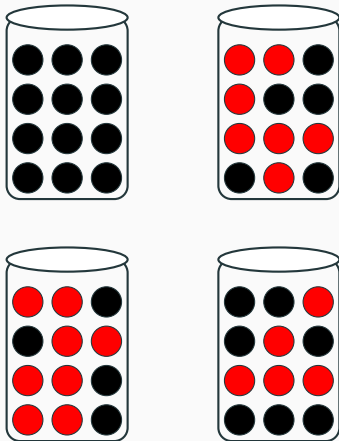
Proposition. $\forall \text{ jar} \in \text{jars}, \exists \text{ ball} \in \text{jar} : \text{ball is red}$



Poll. How can we change the proposition to make it true?

Solution. $\neg \left(\forall \text{ jar} \in \text{jars}, \exists \text{ ball} \in \text{jar} : \text{ball is red} \right)$

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Poll. How can we change the proposition to make it true?

Solution. $\neg \left(\forall \text{ jar} \in \text{jars}, \exists \text{ ball} \in \text{jar} : \text{ball is red} \right)$

That is, $\left(\exists \text{ jar} \in \text{jars}, : \forall \text{ ball} \in \text{jar} : \text{ball is NOT red} \right)$.

Negating quantified statements

Big Picture. To negate quantified statements, “flip” the quantifiers and carry the “not” operator through.

Negating Existential Statements

$$\neg \left(\exists x \in A, \text{assertions about } x \right) \quad \text{is the same as} \\ \forall x \in A, \neg \left(\text{assertions about } x \right)$$

Negating quantified statements

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$\neg(\exists x \in A, \text{assertions about } x)$ is the same as
 $\forall x \in A, \neg(\text{assertions about } x)$

Negating Universal Statements

$\neg(\forall x \in A, \text{assertions about } x)$ is the same as
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Remark. This technique applies even if there are multiple quantifiers (as we saw in the jars slide).

Negating quantified statements

Big Picture. To negative quantified statements, “flip” the quantifiers and carry the “not” operator through.

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Remark. This technique applies even if there are multiple quantifiers (as we saw in the jars slide). E.g. $\neg(\exists \dots : \forall \dots, \exists \dots : \forall \dots, \text{something})$ is the same as $(\forall \dots, \exists \dots : \forall \dots, \exists \dots : \text{NOT something})$.

Outline for Mini-Lecture:

- Using quantified statements
- Proving quantified statements
- Applications

Proving existential statements

Proposition. There is an integer that is even and prime.

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Poll. How to express the proposition in quantified form?

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Proposition (with quantifiers). $\exists x \in \mathbb{Z} : x$ is even and prime.

Poll. How to prove the proposition?

Solution. Consider the integer 2. The integer 2 is both even and prime.
(Here you could possibly justify these properties via the definitions.)

Proving existential statements

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Remark. Proving an existential statement is **akin to finding a counterexample**.

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Remark. Proving an existential statement is **akin to finding a counterexample**.

General strategy.

Proof Template 7 Proving existential statements.

To prove $\exists x \in A$, assertions about x :

Let x be (give an explicit example) ... (Show that x satisfies the assertions.) ... Therefore x satisfies the required assertions. ■

Proving universal statements

Proposition. Every integer that is divisible by 6 is even.

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Proving universal statements

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Proposition (with quantifiers). Let $A = \{x \in \mathbb{Z} : 6 \mid x\}$. Then $\forall x \in A, x$ is even.

Proving universal statements

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Proposition (with quantifiers). Let $A = \{x \in \mathbb{Z} : 6 \mid x\}$. Then $\forall x \in A, x$ is even.

Poll. How to prove the proposition?

Solution. Let $x \in A$. Then there is an integer y such that $x = 6y$.

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Proposition (with quantifiers). Let $A = \{x \in \mathbb{Z} : 6 \mid x\}$. Then $\forall x \in A, x$ is even.

Poll. How to prove the proposition?

Solution. Let $x \in A$. Then there is an integer y such that $x = 6y$. We can rewrite this as $x = 2(3y)$.

Proving universal statements

Proposition. Every integer that is divisible by 6 is even.

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Proposition (with quantifiers). Let $A = \{x \in \mathbb{Z} : 6 \mid x\}$. Then $\forall x \in A, x$ is even.

Poll. How to prove the proposition?

Solution. Let $x \in A$. Then there is an integer y such that $x = 6y$. We can rewrite this as $x = 2(3y)$. Since $3y$ is an integer, x is divisible by 2, and therefore even.

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Remark. Proving an universal statement is **akin to proving an if-then statement**. (“If x is divisible by 6, then x is even”.)

Proving universal statements

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General strategy.

Proof Template 8 Proving universal statements.

To prove $\forall x \in A$, assertions about x :

Let x be any element of A (Show that x satisfies the assertions using only the fact that $x \in A$ and no further assumptions on x .) ... Therefore x satisfies the required assertions.

Puzzle

Evaluate the (partial) student solution below.

Quiz Question (Quantifiers) from Spring 2025

Let $A = \{x \in \mathbb{Z} : 6|x\}$. Prove that $\forall x \in A$, x is even.

(Partial) Student Solution

$$A = \{x \in \mathbb{Z} : 6|x\} \tag{1}$$

$$\exists y : \forall x, x = 6y \tag{2}$$

Puzzle

Evaluate the (partial) student solution below.

Quiz Question (Quantifiers) from Spring 2025

Let $A = \{x \in \mathbb{Z} : 6|x\}$. Prove that $\forall x \in A, x$ is even.

(Partial) Student Solution

$$A = \{x \in \mathbb{Z} : 6|x\} \tag{1}$$

$$\exists y : \forall x, x = 6y \tag{2}$$

Comments

Line 2 in the solution is False.

$$\underbrace{\exists}_\text{there exists} y \underbrace{:}_\text{such that} \underbrace{\forall}_\text{for all} x, x = 6y$$

(No matter what you pick for y , $6y$ can't represent all integers).

Solution Sketch

General Observation

The primary issue (even to students who scored 4/4) was **communication** – that is structuring the mathematical argument clearly.

Solution Using The If-Then Proof Template.

Proposition. Let $A = \{x \in \mathbb{Z} : 6|x\}$. Then $\forall x \in A$, x is even.

Proof.

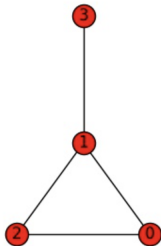
Annotation	Main Text
Convert Prop. to “if-then” form	We show that if $x \in A$, then x is even.
State assumption	Suppose $x \in A$.
Unravel defs.	So $6 x$. That is, there is an integer c such that $x = 6c$.
*** The glue ***	We can write $x = 6c = 2(3c)$.
Unravel defs.	So there is an integer $d(= 3c)$ such that $x = 2d$. In other words, $2 x$.
State conclusion.	So, x is even.

Outline for Mini-Lecture:

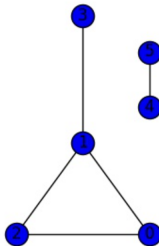
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Application Ex. 1: Graph Theory

Fully connected graph

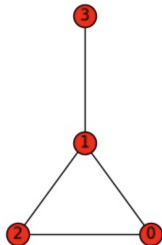


Unconnected graph

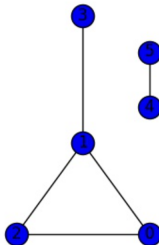


Application Ex. 1: Graph Theory

Fully connected graph



Unconnected graph



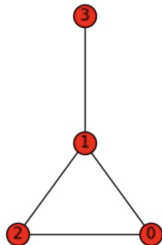
Definition

A graph G is connected if

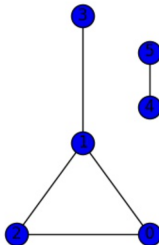
$$\forall u, v \in \text{Vertices}(G), \exists \text{ path } P \text{ from } u \text{ to } v.$$

Application Ex. 1: Graph Theory

Fully connected graph



Unconnected graph



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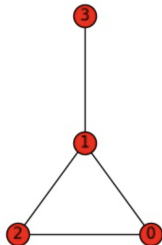
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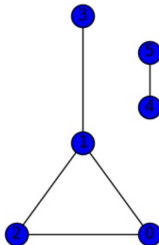
Translation. For all pairs of vertices, there exists a path connecting them.

Application Ex. 1: Graph Theory

Fully connected graph



Unconnected graph



Definition

A graph G is connected if

$$\forall u, v \in \text{Vertices}(G), \exists \text{ path } P \text{ from } u \text{ to } v.$$

Translation. For all pairs of vertices, there exists a path connecting them.

Remark. This combines universal and existential quantifiers naturally.

Application Ex. 2: Randomized Algorithms

Imagine you have a randomized algorithm A that tries to solve a problem P , like generating a valid puzzle or finding a satisfiable solution.

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Imagine you have a randomized algorithm A that tries to solve a problem P , like generating a valid puzzle or finding a satisfiable solution.

You might want to express that with high probability it works for all valid inputs:

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Imagine you have a randomized algorithm A that tries to solve a problem P , like generating a valid puzzle or finding a satisfiable solution.

You might want to express that with high probability it works for all valid inputs:

$$\forall x \in \text{ValidInputs}, \forall \epsilon > 0, \exists N \in \mathbb{N} : \\ \Pr[\text{Algorithm } A \text{ solves } x \text{ in } N \text{ steps}] \geq 1 - \epsilon$$

Translation into words. “For all valid inputs x and for any tiny failure probability ϵ , there exists a number of steps N such that the randomized algorithm succeeds with probability at least $1 - \epsilon$.”

Application Ex. 2: Randomized Algorithms

Imagine you have a randomized algorithm A that tries to solve a problem P , like generating a valid puzzle or finding a satisfiable solution.

You might want to express that with high probability it works for all valid inputs:

$$\forall x \in \text{ValidInputs}, \forall \epsilon > 0, \exists N \in \mathbb{N} : \\ \Pr[\text{Algorithm } A \text{ solves } x \text{ in } N \text{ steps}] \geq 1 - \epsilon$$

Translation into words. “For all valid inputs x and for any tiny failure probability ϵ , there exists a number of steps N such that the randomized algorithm succeeds with probability at least $1 - \epsilon$.”

Application. Finding influential samples in large datasets.

(Woj. et al., 2017, *IEEE Big Data*)



Random group assignments

Ashley, Isabelle: 2

Bailey, Tanna: 3

Blaisdell, Olyver: 4

Brown, Ashton: 13

Brown, Jonathan: 3

Cardenas, Dominic: 12

Cutler, George: 8

Fabik, Roland: 9

Fellows, Audrey: 5

Fitzgerald, Ryan: 4

Fuller, Ryan: 7

Griffen, Clara: 7

Grutsch, Tanner: 13

Hager, Nathan: 11

Horner, Jayden: 1

Johnson, Jaret: 2

Kintzel, Samantha: 1

Lameres, Alexis: 8

Lee, Holden: 6

Mayala, Reece: 5

McLean, Connor: 6

Mcbride, Daniel: 11

Miller, Maverick: 10

Moss, Emily: 4

Mousseau, Matthew: 10

Neuman, Evan: 12

Putnam, John: 6

Ramos, Vydel: 1

Rathbun, Jedidiah: 2

Roberts, Matthew: 3

Roller, Nathan: 9

Ryan, Otis: 7

Shevtsov, Timothy: 11

Sinclair, Aidan: 12

Smith, Charles: 9

Stensland, Emma: 13

Thiessen, Logan: 5

Thompson, Jaiden: 10

Vath, Ethan: 8

Group Exercises: Quantifiers

1. Label each sentence below about the integers as true or false.

- a. $\forall x, \forall y, x + y = 0$.
- b. $\forall x, \exists y : x + y = 0$.
- c. $\exists x : \forall y, x + y = 0$.
- d. $\exists x, \exists y : x + y = 0$.
- e. $\forall x, \forall y, xy = 0$.
- f. $\forall x, \exists y : xy = 0$.
- g. $\exists x : \forall y, xy = 0$.
- h. $\exists x, \exists y : xy = 0$.

2. The notation $\exists!$ can be read "there is a unique." Label each sentence below true or false.

- a. $\exists! x \in \mathbb{N} : x^2 = 4$.
- b. $\exists! x \in \mathbb{Z} : x^2 = 4$.
- c. $\exists! x \in \mathbb{N} : x^2 = 3$.
- d. $\exists! x \in \mathbb{Z} : \forall y \in \mathbb{Z}, xy = x$.
- e. $\exists! x \in \mathbb{Z} : \forall y \in \mathbb{Z}, xy = y$.

3. For each sentence below, write the negation of the sentence, but place the $\neg$ symbol as far to the right as possible. Then rewrite the negation in English. The first problem is done for you.

- a. $\forall x \in \mathbb{Z}, x$ is odd. **Solution:**
 $\exists x \in \mathbb{Z} : \neg(x \text{ is odd.})$. In
English: "There is an
integer which is not odd."

- b. $\forall x \in \mathbb{Z}, x < 0$.
- c. $\exists x \in \mathbb{Z} : x = x + 1$.
- d. $\exists x \in \mathbb{N} : x > 10$.
- e. $\forall x \in \mathbb{N}, x + x = 2x$.
- f. $\exists x \in \mathbb{Z} : \forall y \in \mathbb{Z}, x > y$.
- g. $\forall x \in \mathbb{Z}, \forall y \in \mathbb{Z}, x = y$.
- h. $\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z} : x + y = 0$.

Solutions to Group Exercises #1

Label each sentence below about the integers as true or false.

- a. $\forall x, \forall y, x + y = 0$. **False**. (Counter-example: $x = 1, y = 1$.)
- b. $\forall x, \exists y : x + y = 0$. **True**. (Set $y = -x$.)
- c. $\exists x : \forall y, x + y = 0$. **False**. (Counter-example: If $x = 0$, set $y = 1$. If $x \neq 0$, set $y = 0$.)
- d. $\exists x, \exists y : x + y = 0$. **True**. (Set $x = 1, y = -1$.)
- e. $\forall x, \forall y, xy = 0$. **False**. (Counterexample: set $x = y = 1$.)
- f. $\forall x, \exists y : xy = 0$. **True**. (Set $y = 0$.)
- g. $\exists x : \forall y, xy = 0$. **True**. (Set $x = 0$.)
- h. $\exists x, \exists y : xy = 0$. **True**. (Set $x = y = 0$.)

Solutions to Group Exercises #2

The notation $\exists!$ can be read "there is a unique." Label each sentence below true or false.

- a. $\exists! x \in \mathbb{N} : x^2 = 4$. **True.** ($x = 2$ is the only solution to the equation in the natural numbers.)
- b. $\exists! x \in \mathbb{Z} : x^2 = 4$. **False.** ($x = 2$ and $x = -2$ are two different integers which solve the equation.)
- c. $\exists! x \in \mathbb{N} : x^2 = 3$. **False.** (There is no solution to the equation in the set of natural numbers.)
- d. $\exists! x \in \mathbb{Z} : \forall y \in \mathbb{Z}, xy = x$. **True.** ($x = 0$ is the only solution to the equation in the integers.)
- e. $\exists! x \in \mathbb{Z} : \forall y \in \mathbb{Z}, xy = y$. **True.** ($x = 1$ is the only solution to the equation in the integers.)

Solutions to Group Exercises #3 a-d

For each sentence below, write the negation of the sentence, but place the $\neg$ symbol as far to the right as possible. Then rewrite the negation in English.

- a. $\forall x \in \mathbb{Z}, x$ is odd. Solution: $\exists x \in \mathbb{Z} : \neg(x \text{ is odd})$. In English: "There is an integer which is not odd." Alternatively, we could write the negated proposition as $\exists x \in \mathbb{Z} : (x \text{ is even})$. In English: "There is an integer which is even." Note: the negated proposition is **True**.
- b. $\forall x \in \mathbb{Z}, x < 0$. Solution: $\exists x \in \mathbb{Z} : \neg(x < 0)$. In English: "There is an integer which is not less than zero." Alternatively, we could write $\exists x \in \mathbb{Z} : x \geq 0$. In English: "There is an integer which is greater than or equal to zero." Note: the negated proposition is **True**.
- c. $\exists x \in \mathbb{Z} : x = x + 1$. Solution: $\forall x \in \mathbb{Z} : \neg(x = x + 1)$. That is, $\forall x \in \mathbb{Z} : (x \neq x + 1)$. In English: "No integer equals one plus itself." Note: the negated proposition is **True**.
- d. $\exists x \in \mathbb{N} : x > 10$. Solution: $\forall x \in \mathbb{N} : \neg(x > 10)$. Alternatively, $\forall x \in \mathbb{N} : x \leq 10$. In English: "Each natural number is less than or equal to 10." Note: the negated proposition is **False**.

Solutions to Group Exercises #3 e-h

For each sentence below, write the negation of the sentence, but place the $\neg$ symbol as far to the right as possible. Then rewrite the negation in English.

- e. $\forall x \in \mathbb{N}, x + x = 2x$. Solution: $\exists x \in \mathbb{N} : \neg(x + x = 2x)$. That is, $\exists x \in \mathbb{N} : x + x \neq 2x$. In English: "There is a natural number such that the sum of the number with itself does not equal double itself." Note: the negated proposition is **False**.
- f. $\exists x \in \mathbb{Z} : \forall y \in \mathbb{Z}, x > y$. Solution: $\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z} : \neg(x > y)$. That is, $\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z} : x \leq y$. In English: "For all integers, there is another integer which is at least as large as it." Note: the negated proposition is **True**.
- g. $\forall x \in \mathbb{Z}, \forall y \in \mathbb{Z}, x = y$. Solution: $\exists x \in \mathbb{Z}, \exists y \in \mathbb{Z} : \neg(x = y)$. That is, $\exists x \in \mathbb{Z}, \exists y \in \mathbb{Z} : x \neq y$. In English: "There are two integers which do not equal each other." Note: the negated proposition is **True**.
- h. $\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z} : x + y = 0$. Solution: $\exists x \in \mathbb{Z} : \forall y \in \mathbb{Z} : \neg(x + y = 0)$. That is, $\exists x \in \mathbb{Z} : \forall y \in \mathbb{Z} : (x + y \neq 0)$. In English: "There is an integer such that the sum with all other integers does not equal zero." Note: the negated proposition is **False**.